

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SOFTWARE REQUIREMENTS SPECIFICATION
(SRS)

Project Title: Valerune — Web-Based Retail Inventory Management System

Standard Academic Compliance: IEEE Std 830-1998
B.Tech 3rd Year Capstone Project Submission

Attribute	Details
Project Title	Valerune — Web-Based Retail Inventory Management System
Student Name	1. Student Developer (Roll No: [Roll Number]) 2. Student Developer (Roll No: [Roll Number])
Degree & Program	Bachelor of Technology (B.Tech in CSE)
Academic Year	2026 – 2027
Department	Department of Computer Science & Engineering
Project Guide	Prof. / Dr. [Project Guide Name] (Assistant Professor)
Status & Submission	3rd Year Project Specification Baseline October 2026

Document Revision & Approval History

Version	Date	Description / Major Changes	Prepared By	Reviewed By
0.1	15-Sep-2026	Initial Architecture Draft & Scope Definition	Engineering Team	Project Guide
1.0	28-Sep-2026	Added Warehouse Management & Supplier CRUD API Modules	Engineering Team	Project Guide
1.2	30-Sep-2026	Integrated JWT Security Filter & Order Fulfillment Workflow	Engineering Team	Project Guide
2.0	01-Oct-2026	Final 3rd Year Mid-Term SRS Baseline (IEEE Std 830-1998)	Engineering Team	HOD / Committee

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, clear, and formal baseline of system requirements for Valerune — Web-Based Retail Inventory Management System. It defines the functional and non-functional specifications necessary for academic evaluation, architectural review, and software maintenance.

1.2 Scope of the System

Valerune is designed to solve retail supply chain inefficiencies by delivering an end-to-end multi-tier web platform that integrates customer e-commerce interactions with administrative multi-warehouse stock management, supplier tracking, and order fulfillment.

In-Scope Capabilities: Customer Storefront, Stateless JWT Security with RBAC (CUSTOMER, ADMIN), Product Catalog CRUD, Multi-Warehouse Inventory Allocation across 4 regional hubs (WH-NORTH, WH-CENTRAL, WH-WEST, WH-SOUTH), Stock Movement Audit Logging, Order Fulfillment State Transitions (PENDING -> PROCESSING -> COMPLETED / CANCELLED), Supplier Directory Management, Operational KPI Analytics Dashboard.

Out-of-Scope Boundaries: Real-world credit card payment gateway terminals, physical hardware barcode/RFID scanner devices, and 3PL courier tracking API integrations.

1.3 Definitions, Acronyms, and Abbreviations

Category	Term / Acronym	Definition / Standard Context
Standard	SRS	Software Requirements Specification (IEEE Std 830-1998 format).
Architecture	API	Application Programming Interface for data interchange via JSON/HTTPS.
Security	JWT / RBAC	JSON Web Token for stateless authorization; Role-Based Access Control.
Database	CRUD	Create, Read, Update, Delete basic data manipulation operations.
Network	REST	Representational State Transfer stateless software architectural style.
ORM	JPA / Hibernate	Java Persistence API / Object-Relational Mapping for relational DB entities.
Domain	SKU	Stock Keeping Unit unique product identification code.

2. Overall Description

2.1 Product Perspective

Valerune operates in a client-server paradigm:
React 19 / TypeScript SPA Frontend <--> REST HTTP/JSON + Bearer JWT <--> Spring Boot 3.3 Backend Services <--> Spring Data JPA / Hibernate ORM <--> PostgreSQL 17 Database.

2.2 User Classes and Characteristics

User Role / Persona	Technical Proficiency	Core Responsibilities & Rights
System Administrator	High (DevOps / Inventory Mgmt)	Full access: Product catalog CRUD, warehouse stock adjustments, supplier directory management, order fulfillment, and stock audit logs.
Registered End User	Intermediate (General Web User)	Authenticated profile management, shopping cart, wishlist, checkout order placement, and customer order history tracking.
Guest / Public Visitor	Basic	Read-only access to public storefront catalog, product detail pages, search, multi-category filtering, and self-registration.

3. System Features and Functional Requirements

3.1 Module 1: Authentication and Access Governance

Req ID	Feature Description	Input / Validation Rule	Priority
FR-1.1	User Account Registration	Email, Name, Password (>=6 chars). Defaults to CUSTOMER role.	High
FR-1.2	Secure Credential Login	Registered Email + Password; verifies BCrypt hash and returns JWT token.	High
FR-1.3	BCrypt Password Hashing	Hashes raw passwords using BCrypt algorithm prior to DB persistence.	High
FR-1.4	Stateless JWT Issuance	Generates signed 512-bit HMAC JWT token upon successful authentication.	High
FR-1.5	Role-Based Route Guard	Restricts /admin frontend routes and admin REST APIs to ADMIN role.	High
FR-1.6	Get Current User Profile	Extracts user profile using Bearer JWT token from Authorization header.	Medium
FR-1.7	User Session Logout	Clears SecurityContextHolder and client localStorage JWT token.	Medium

3.2 Module 2: Core Processing & Business Operations

Req ID	Feature Description	Input / Validation Rule	Priority
FR-2.1	Create Product Entry	Requires unique SKU (VAL-***), non-empty name, price > 0, category.	High
FR-2.2	Retrieve Retail Catalog	Returns list of active products with price, rating, category, image URL.	High
FR-2.3	Product Detail Lookup	Fetches specific product record by UUID string.	High
FR-2.4	Update Product Profile	Modifies product attributes while preserving SKU uniqueness.	Medium
FR-2.5	Soft Delete Product	Sets active flag to false without deleting historical transaction data.	Medium
FR-2.6	Search & Filter Catalog	Filters catalog by keyword text, category, price range, and sorting.	High
FR-3.1	Register Warehouse Hub	Requires unique code (WH-***), name, location, capacity > 0.	High
FR-4.1	Create Stock Mapping	Maps product UUID to warehouse code with initial quantity >= 0.	High
FR-4.3	Stock Level Adjustment	Updates quantity on hand and reserved with non-negative validation.	High
FR-4.4	Low Stock Alerting	Triggers warning whenever quantity on hand <= reorder level.	High
FR-4.5	Optimistic Locking Control	Enforces version field check to prevent concurrent stock overwrite errors.	High

3.3 Module 3: Reporting, Visualization & Administrative Control

Req ID	Feature Description	Input / Validation Rule	Priority
FR-5.1	Log Movement Audit Event	Records stock transaction (INBOUND, OUTBOUND, ADJUSTMENT, RESERVATION).	High
FR-6.1	Create Sales Order	Accepts order payload with line items, total calculation, and warehouse code.	High
FR-6.2	Inventory Reservation	Automatically deducts quantity from on-hand and reserves stock.	High
FR-6.4	Order Fulfillment Workflow	Transitions order status (PENDING -> PROCESSING -> COMPLETED / CANCELLED).	High
FR-7.1	Register Vendor Supplier	Requires unique supplier code, company name, contact name, email, phone.	Medium
FR-8.1	Operations Dashboard Metrics	Aggregates total products, total stock, low-stock count, active orders.	Medium
FR-8.3	DataInitializer Seeding	Seeds 50 Valerune retail products and 4 warehouse records on startup.	High

4. External Interface Requirements

- User Interface: Customer Storefront Portal & Admin Operational Dashboard.
- Software Interfaces: React 19 SPA, Spring Boot 3.3 REST API, PostgreSQL 17 DB, Hibernate ORM.
- Communication Interfaces: HTTP/HTTPS REST JSON with Bearer JWT headers.

5. Non-Functional Requirements (NFRs)

Attribute	Target Metric / Standard	Verification & Testing Method
5.1 Performance	Page load latency < 1.5s; REST API response time < 350ms for query endpoints.	Postman & Spring Boot Actuator health/metrics endpoints.
5.2 Security	Zero plain-text passwords (BCrypt hashed); 512-bit HMAC JWT tokens; SQL injection immunity via JPA parameterized queries.	Security code review & JUnit spring-security test suite.
5.3 Reliability	System uptime >= 99% during demonstration; database transaction rollback on runtime exceptions.	JUnit 5 integration test suite (126 tests passing).
5.4 Portability	Cross-browser execution (Chrome, Edge, Firefox, Safari); fully responsive viewport (mobile to desktop).	Responsive viewport inspection in Developer Tools across devices.
5.5 Maintainability	Layered Spring Boot architecture (Controller-Service-Repository); modular React TypeScript components.	Static code audit & TypeScript compiler verification.

6. System Design and Analysis Models

Relational Database Schema (3NF): 8 primary database entities (`users`, `products`, `warehouses`, `inventories`, `suppliers`, `orders`, `order\_items`, `stock\_movements`).

7. Academic Review & Sign-Off

Project Coordinator / Guide	Internal Examiner	Head of Department (HOD)
Signature: _____ Date: _____/_____/2026	Signature: _____ Date: _____/_____/2026	Signature: _____ Date: _____/_____/2026